

NETWORKS AND COMPLEXITY

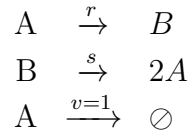
Solution 14-10

This is an example solution from the forthcoming book Networks and Complexity.

Find more exercises at <https://github.com/NC-Book/NCB>

Ex 14.10: Glycolysis [Lvl. 4]

Glycolysis is a process by which your cells turn sugar into energy. The essence of glycolysis can be captured in the following conceptual model:



Here, A describes the amount of available energy. Energy is needed to start a process (first reaction), which then produces more energy (second reaction), which is used to sustain the organism (third reaction).

- a) Let a be the amount of A and b the amount of B in the system. Then, write the ODE system that captures the dynamics of these variables, and show that there's no steady state in which $a^* > 0$ except in a special case.

Solution

Using the mass action laws, we write

$$\dot{a} = 2sb - (r + 1)a \quad (1)$$

$$\dot{b} = ar - sb \quad (2)$$

This is a linear system and every linear system only has exactly one steady state where all variables are zero. To show this explicitly, we compute the steady state. From the second equation, we get the steady state condition

$$b = \frac{r}{s}a \quad (3)$$

and substituting this into the first differential equation gives us the condition

$$0 = 2ra - (r + 1)a = (r - 1)a \quad (4)$$

So, except in the exceptional (i.e. degenerate) state, where the parameters are chosen such that $r = 1$, the only steady state is $a = 0$.

- b) The absence of a robust steady state where we have a constant level of stored energy is a problem. Let's assume that $r = a^p$, where $p \neq 0$ is a new parameter. Show that there is now a steady state with $a^* > 0$.

Solution

We can proceed as before until we reach the steady state condition

$$0 = (r - 1)a. \quad (5)$$

We are no longer interested in steady states where $a = 0$, therefore we can now divide by a , which leads us to the familiar condition

$$r = 1 \quad (6)$$

but now we use $r = a^p$ and hence

$$a^p = 1 \quad (7)$$

and taking the p 's root on both sides

$$a = \sqrt[p]{1} = 1. \quad (8)$$

When dealing with roots it's good to remember that there can be multiple solutions, but the root of a positive number only ever has one positive real-valued solution, and as a represents the concentration of substance (ATP), it should be positive and real. So, unless $p = 0$ there is a steady state in which we have a constant level of energy available in the cell.

- c) Show that the steady state is unstable for $p > 0$. (This leads up to a nice biological insight, check the solution).

Solution

When computing the Jacobian, we have to keep in mind that $r = a^p$ such that

$$\dot{a} = 2sb - a^{p+1} - a \quad (9)$$

$$\dot{b} = a^{p+1} - sb \quad (10)$$

We compute the derivatives

$$J_{11} = \left. \frac{\partial}{\partial a} 2sb - a^{p+1} - a \right|_* = \left. -(p+1)a^p - 1 \right|_* = -(p+1) - 1 = -(p+2) \quad (11)$$

$$J_{12} = \left. \frac{\partial}{\partial b} 2sb - a^{p+1} - a \right|_* = 2s \quad (12)$$

$$J_{21} = \left. \frac{\partial}{\partial a} a^{p+1} - sb \right|_* = p+1 \quad (13)$$

$$J_{22} = \left. \frac{\partial}{\partial b} a^{p+1} - sb \right|_* = -s \quad (14)$$

Hence, the Jacobian is

$$\mathbf{J} = \begin{pmatrix} -(p+2) & 2s \\ p+1 & -s \end{pmatrix}. \quad (15)$$

We compute the characteristic polynomial

$$\left| \begin{pmatrix} -(p+2) - \lambda & 2s \\ p+1 & -s - \lambda \end{pmatrix} \right| = \lambda^2 + (p+2+s)\lambda + s(p+2) - 2s(p+1) \quad (16)$$

$$= \lambda^2 + (p+2+s)\lambda - sp \quad (17)$$

and solve

$$0 = \lambda^2 + (p + 2 + s)\lambda - sp \quad (18)$$

$$0 = (\lambda + (p + 2 + s)/2)^2 - (p + 2 + s)^2/4 - sp \quad (19)$$

$$(\lambda + (p + 2 + s)/2)^2 = (p + 2 + s)^2/4 + sp \quad (20)$$

$$\lambda + (p + 2 + s)/2 = \pm \sqrt{(p + 2 + s)^2/4 + sp} \quad (21)$$

$$(22)$$

So the leading eigenvalue is

$$\lambda_1 = -\frac{p + 2 + s}{2} + \sqrt{\left(\frac{p + 2 + s}{2}\right)^2 + sp}. \quad (23)$$

for $p > 0$ the expression under the square root is positive. Hence, the eigenvalues are real. Thus, the system is stable if (and only if)

$$\frac{p + 2 + s}{2} > \sqrt{\left(\frac{p + 2 + s}{2}\right)^2 + sp} \quad (24)$$

For $p > 0$ this condition is always false. If you don't see this immediately consider that both sides of the inequality are positive for $p > 0$, so we can square both sides

$$\left(\frac{p + 2 + s}{2}\right)^2 > \left(\frac{p + 2 + s}{2}\right)^2 + sp \quad (25)$$

$$0 > sp \quad (26)$$

which isn't possible if $p > 0$, since s is a rate, so $s > 0$.

So p needs to be negative for a stable steady state in which the cell has a positive supply of energy. That means the enzyme that catalyzes the $A \rightarrow B$ step must act more slowly if there is a lot of A around. In biochemical terms, we can say it is inhibited by its own substrate. This 'substrate inhibition' is quite rare in enzymes, but our model suggests that substrate inhibition is needed to keep glycolysis stable. Analyzing a simple model has given us a biological hypothesis that could be tested in experiments. If we did, we would find it to be true (with $p \approx -1$)